

Numerical Methods for initial value problems

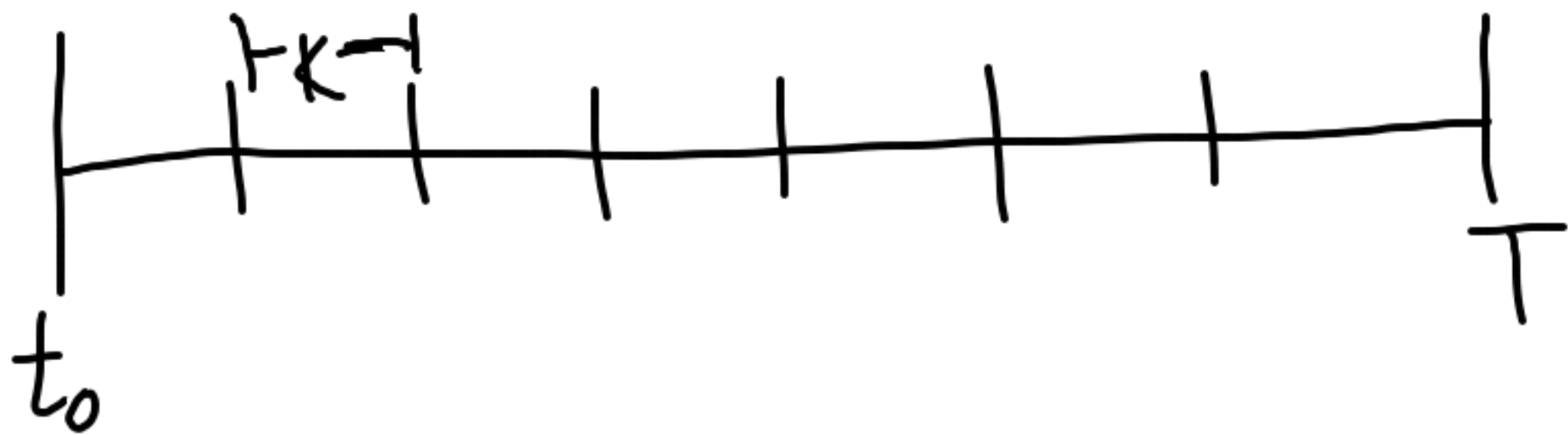
Computational work:

$$N \times (\text{work per step})$$

$$= \frac{T-t_0}{k} \times (\text{work per step})$$

We will compute values
 $U^1, U^2, \dots$ in sequence.

$$N = \frac{T-t_0}{k}$$



Some basic methods

Explicit Euler

$$U^{n+1} = U^n + k f(U^n)$$

$$\frac{U^{n+1} - U^n}{k} = f(U^n)$$

Implicit Euler

$$U^{n+1} = U^n + k f(U^{n+1})$$

(Need to solve algebraic eqns. at each step)

Trapezoidal method

$$U^{n+1} = U^n + \frac{k}{2} (f(U^n) + f(U^{n+1}))$$

Implicit midpoint method

$$U^{n+1} = U^n + k f\left(\frac{U^n + U^{n+1}}{2}\right)$$

Leapfrog method

$$U^{n+1} = U^{n-1} + 2k f(U^n)$$

Truncation error analysis

$$\frac{U^{n+1} - U^{n-1}}{2k} = f(U^n)$$

$$\frac{u(t_n+k) - u(t_n-k)}{2k} = f(u(t_n)) + \tau^n$$

$$u(t_n) + k u'(t_n) + \frac{k^2}{2} u''(t_n) - \left(u(t_n) - k u'(t_n) + \frac{k^2}{2} u''(t_n) \right)$$

$$\frac{\cancel{u(t_n)} + ku'(t_n) + \frac{k^2}{2}\cancel{u''(t_n)} + \frac{k^3}{6}u'''(t_n) - (\cancel{u(t_n)} - ku'(t_n) + \frac{k^2}{2}\cancel{u''(t_n)} - \frac{k^3}{6}u'''(t_n)) + O(k^5)}{2k} = u'(t_n) + \tau^n$$

$$\cancel{u'(t_n)} + \frac{k^2}{6}u'''(t_n) + O(k^4) = \cancel{u'(t_n)} + \tau^n$$

$$\tau^n = \frac{k^2}{6}u'''(t_n) + O(k^4) \quad (\text{LTE})$$

We call $\rho = k\tau$ the
"one-step error."

$$f(u(t_n)) = u'(t_n)$$

How to achieve higher
order accuracy

- ① Use more derivatives *multiderivative*
- ② Use more f evaluations *multistage*
(Runge-Kutta)
- ③ Use more previous
Steps *Multistep*

① Multiderivative

$$u(t_n + k) = u(t_n) + \underbrace{k u'(t_n)}_{f(u(t_n))} + \frac{k^2}{2} u''(t_n) + \frac{k^3}{6} u'''(t_n) + \dots$$

$$u_i''(t_n) = \frac{d}{dt} u_i'(t_n) = \frac{d}{dt} f_i(u(t_n)) = \sum_j \frac{\partial f_i}{\partial u_j} \frac{du_j}{dt}$$

$$u''(t_n) = f' f$$

$$u_i'''(t_n) = \frac{d}{dt} \left(\sum_j \frac{\partial f_i}{\partial u_j} f_j \right) = \sum_j \sum_l \frac{\partial^2 f_i}{\partial u_j \partial u_l} f_j f_l + \sum_j \frac{\partial f_i}{\partial u_j} \sum_l \frac{\partial f_j}{\partial u_l} f_l$$

$$u'''(t_n) = f''(f, f) + f' f' f$$

The # of terms grows rapidly.

But we can use automatic differentiation.
(e.g. Jax)

② Multistage (Runge-Kutta)

Example:

$$U^* = U^n + \frac{k}{2} f(U^n)$$

$$U^{n+1} = U^n + k f(U^*)$$

In general: $Y_i = U^n + k \sum_{j=1}^s a_{ij} f(Y_j)$

$$U^{n+1} = U^n + k \sum_{j=1}^s b_j f(Y_j)$$

Y_j : Stages
 s : # of stages

We can denote a specific method by

	A
b	

"Butcher tableau"

0	0
1/2	0
0	1

$$Y_1 = U^n$$

$$Y_2 = U^n + \frac{k}{2} f(Y_1)$$

$$U^{n+1} = U^n + k f(Y_2)$$

0	0	0	0
1/2	0	0	0
0	1	0	0
0	0	1	0
1/6	1/3	1/3	1/6

"The RK method"

4th-order accurate

③ Multistep methods

Example: $U^{n+1} = U^{n-1} + 2kf(U^n)$



In general:

$$\sum_{j=0}^r \alpha_j U^{n+j} = k \sum_{j=0}^r \beta_j f(U^{n+j})$$

RK

+ Self-starting
+ Easy to adapt k

LM

+ Only evaluate f
once per step